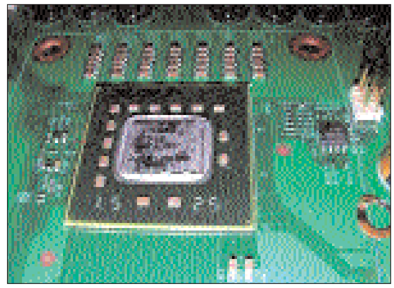
24 million units by the end of its life, running second to Sony's behemoth and overtaking Nintendo's Gamecube by a small, but significant margin. Mission accomplished, the Xbox brand was etched into the psyche of console gamers, rap artists, and PC gamers tired of the upgrade-and-patch dance of PC games.

Almost immediately after releasing the Xbox, Microsoft got to work on its successor. The almost four years of head-start that the 360 would thus get, was instrumental in the design that the Xbox 360 now sports-inside and out. It also allowed Microsoft to correct a major drawback that the original of the species had: Microsoft did not own the IP to any of the Xbox's internals—Intel owned the processor, NIVIDIA the graphics chip. Microsoft thus had to rely on two third-parties to supply essential elements of the console, and was dependent on their whims and manufacturing capabilities to cut prices or throttle production as needed. With the Xbox 360, Microsoft owns both the processor and the graphics chip, and has the flexibility to set up fabrication plants, drop prices, and shrink processors as and when needed.



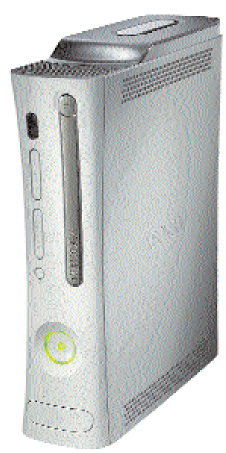
The Xenon processor is a three core unit designed by IBM

Original Of The Species

The Xbox 360 is unique in several aspects. The most benign of which is that the 360 is the only console to simultaneously launch across the three major gaming markets of USA, Europe and Japan. More interesting is the unit's hardware design. It seems to do a fine job at balancing raw power, flexibility and ease-of-use. It is also the potential proving ground for some new school of thought, schools rather—be it unified memory architecture, procedural rendering or unified shader architecture. The 360 is also the first gaming console which pays attention to its software user interface. It is a far cry from the ugly brick which was its predecessor—the unit has a pleasant white exterior, a concave body, a chrome DVD tray, and the ring of light—a circle of LEDs which indicate wireless controller and unit status.

The cool new Xbox controller

Let us begin this dissection from within, from the core of what makes the 360 so unique.



The sleek looking Xbox360

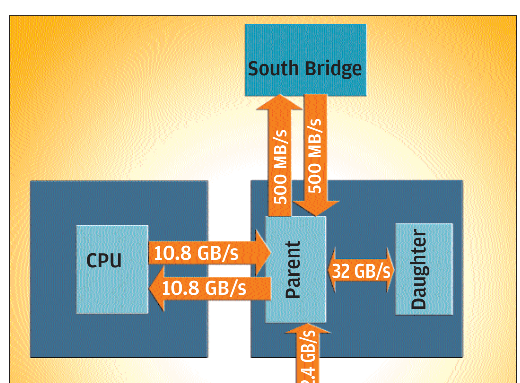
A Procedural World

Microsoft approached once-rival IBM to design the processor of the Xbox 360. The resulting processor is a three core unit for which IBM borrowed design ideas from the Broadband Processor Architecture (BPA) that produced the PlayStation 3's Cell processor. The 360's processor, codenamed Xenon consists of three identical cores to one side, each with 64 KB of L1 cache (32K for data and 32K for instructions) and a large 1 MB of L2 cache which is shared by all three cores, on the other side. Each of the cores has two hardware threads for simultaneous multithreading (SMT); the entire processor can thus process a total of six hardware threads simultaneously. This multithreaded approach is vital to keeping the parts constantly fed—something that increases processor and power efficiency as well as parallelising disparate tasks across the processor cores.

Traditionally, cache is meant to act as a holding unit for frequently used data, such that the processor need not travel all the way to the system's main memory to fetch and process information which is often used. Under the Xenon however, the 1 MB of L2 cache plays a much more interesting and flexible role: it is possible to use the cache in the traditional sense but it can also be shared between the processor and the graphics chip of the Xbox 360. This is a very unique architecture which allows a programmer to push content from the L2 cache to the GPU, much like a FIFO queue (First In, First Out). The L2 can thus be used as a temporary storage for the GPU.

Before we head further into the 360's core, let's take a look at one of the new "schools of thought" which we alluded to earlier in this piece. If you have followed Will Wright's upcoming PC game called Spore, you are probably aware of the concept of procedural content. The idea is simple: free the artists from producing high-resolution art assets and content, and utilise hardware and software to churn out the same. A typical game is bound to contain repeated content-in a title such as Grand Theft Auto III, for example, you will see trees, lampposts, fire hydrants, trash bins and so on, being used throughout the game. Presently, a game uses the same model and texture for each of these common objects and repeats them throughout the title. The result is a bland, uniform look to a game, which reduces the immersion.

Procedural creation of content would take a template of an object, run it through a processor and churn out a finished product in real-time. Traditionally, a game developer would have to hire a team of artists to produce say, trees, then put them on disk, feed them to the computer memory and then put them out to the graphics chip for rendering. With a procedural approach, the developer would only need to define the tree mathematically and ship this information on



The bandwidth available between the core elements of the Xbox 360